

ROBINSON, IL, USA

WMO: 720319

Lat: 39.016N

Lon: 87.650W

Elev: 141

StdP: 99.64

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-12.3	-19.8	0.7	-14.2	-17.1	0.8	-11.3	11.7	5.1	10.8	2.6	2.4	310	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	33.8	24.2	32.6	23.8	31.4	23.3	26.5	31.3	25.6	30.2	24.9	29.2	3.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.7	29.2	24.3	19.6	28.4	23.6	18.7	27.4	83.4	31.4	79.7	29.8	76.2	29.3	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.7	7.7	DB	-19.7	35.5	3.4	1.6	-22.1	36.6	-24.1	37.6	-26.1	38.5	-28.6	39.6	(4)
(5)			WB	-20.1	27.8	3.2	0.8	-22.4	28.4	-24.3	28.9	-26.1	29.3	-28.4	29.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.7	-0.9	0.6	7.1	13.1	18.9	23.5	24.2	23.3	20.2	13.6	6.7	1.7
	DBStd	10.28	6.38	6.57	6.25	5.15	4.73	3.14	2.90	2.89	3.64	5.06	5.25	5.58
	HDD10.0	1180	342	271	132	27	1	0	0	0	0	19	126	262
	HDD18.3	2724	597	497	350	168	54	2	1	3	22	164	351	515
	CDD10.0	2171	3	7	43	119	276	403	439	412	306	132	26	6
	CDD18.3	674	0	0	3	11	70	156	181	156	78	18	0	0
	CDH23.3	6653	0	0	23	121	656	1504	1801	1506	842	196	3	0
	CDH26.7	2514	0	0	2	19	214	602	732	575	313	57	0	0
Wind	WSAvg	3.1	4.1	3.9	4.0	3.9	3.2	2.6	2.0	1.7	1.9	2.8	3.3	3.6
Precipitation	PrecAvg	1042	66	59	93	98	115	92	105	90	81	74	91	76
	PrecMax	1398	240	142	239	243	264	161	274	199	229	213	290	205
	PrecMin	725	5	11	17	29	30	17	21	24	2	13	11	5
	PrecStd	178	55	31	51	47	64	40	56	47	62	49	50	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.3	19.8	26.3	29.1	32.8	35.0	37.6	35.0	33.9	31.2	23.5	18.6
		MCWB	14.6	14.8	16.9	17.9	21.8	23.5	24.0	25.0	22.0	21.3	15.3	16.0
	2%	DB	14.1	16.6	22.8	26.4	31.0	33.2	33.6	32.9	32.2	28.0	20.5	15.3
		MCWB	11.8	13.6	15.9	17.5	20.7	23.9	24.9	24.3	22.5	19.4	15.9	13.6
	5%	DB	11.4	13.9	19.9	24.6	29.1	32.1	32.2	31.6	30.2	25.2	17.7	12.7
		MCWB	9.5	11.1	14.5	16.6	20.5	23.1	25.0	24.1	22.2	17.9	13.7	10.8
	10%	DB	8.5	10.5	17.4	22.3	27.3	30.3	30.7	30.0	28.3	22.6	15.4	10.2
		MCWB	6.7	7.6	13.5	15.8	19.7	22.1	24.2	23.5	20.8	16.8	11.3	8.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	16.3	18.3	20.9	23.8	26.7	27.8	27.8	25.6	22.9	18.7	16.7
		MCDB	16.7	18.7	23.4	25.1	29.4	32.3	32.1	32.7	29.8	29.2	21.8	18.2
	2%	WB	12.5	14.2	17.0	19.2	22.8	25.6	26.7	26.3	24.6	21.4	16.4	13.9
		MCDB	13.9	16.1	21.5	23.3	28.2	30.9	31.2	30.7	29.2	26.0	18.9	15.0
	5%	WB	9.9	11.6	15.5	17.9	21.9	24.5	25.8	25.4	23.4	19.9	14.4	11.3
		MCDB	11.0	13.6	19.4	22.1	26.8	29.5	30.2	29.7	28.1	23.2	17.4	12.4
	10%	WB	6.4	7.7	13.8	16.8	20.9	23.6	25.0	24.5	22.3	18.1	11.9	8.3
		MCDB	8.0	10.3	16.6	21.1	25.6	28.2	29.2	28.2	26.4	21.6	14.4	9.8
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.4	10.5	11.9	11.2	11.4	11.3	12.1	13.8	12.7	10.7	8.2
		MCDBR	11.8	13.4	13.9	14.9	13.5	13.2	12.7	13.6	15.7	15.8	14.2	11.2
		MCWBR	10.0	10.8	8.3	7.7	5.5	5.0	5.4	6.3	7.1	7.9	9.3	9.6
	5% WB	MCDBR	10.9	12.6	12.5	11.8	11.6	11.1	10.7	11.1	12.6	12.8	12.5	10.4
		MCWBR	10.1	10.9	8.1	7.5	5.6	4.9	5.5	6.2	6.6	7.2	9.6	9.7
Clear-Sky Solar Irradiance	taub		0.310	0.321	0.357	0.408	0.444	0.448	0.467	0.476	0.426	0.363	0.322	0.305
	taud		2.445	2.399	2.371	2.229	2.180	2.203	2.150	2.110	2.236	2.413	2.477	2.470
	Ebn at Noon		864	902	900	869	841	835	815	796	816	842	843	842
	Edh at Noon		81	97	111	137	147	143	150	152	126	96	79	74
All-Sky Solar Radiation	RadAvg		1.86	2.65	3.70	4.84	5.47	6.30	6.11	5.57	4.64	3.31	2.23	1.59
	RadStd		0.21	0.29	0.33	0.30	0.48	0.40	0.38	0.33	0.33	0.34	0.24	0.16

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	N/A

Nomenclature: See separate page